

*“SENSE OF TASTE ITS
RECEPTOR AND ITS
PATHWAYS”*

LEARNING OBJECTIVE

At the end of lecture student must be able to

- ***Define taste***
- ***Describe taste buds and its structure***
- ***Explain taste sensations,***
- ***Analyze the different types of papillae***
- ***Understand Taste pathway.***

Sense of taste

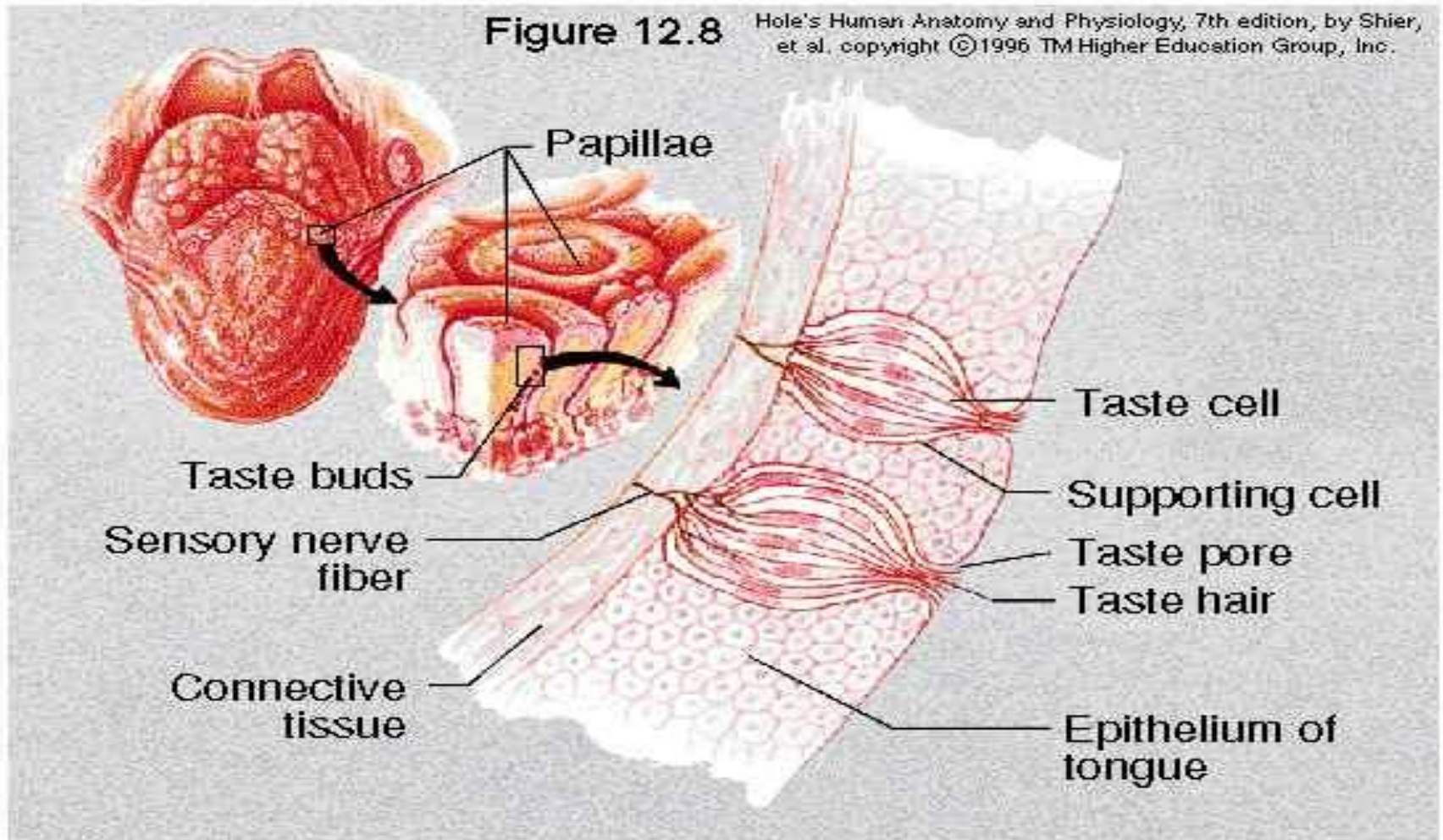
- **Taste (gustation)** is a form of direct chemoreception, and is one of the traditional five senses.
- It refers to the ability to detect the flavor of substances such as food, certain minerals, and poisons, etc
- The sensation of taste is classified into basic tastes:
 - sweetness
 - bitterness
 - Sourness
 - saltiness
 - Umami (savory)

Taste receptor

- A Taste receptor is a type of receptor which facilitates the sensation of taste.
- They are located on type III cells of taste buds.

Taste buds

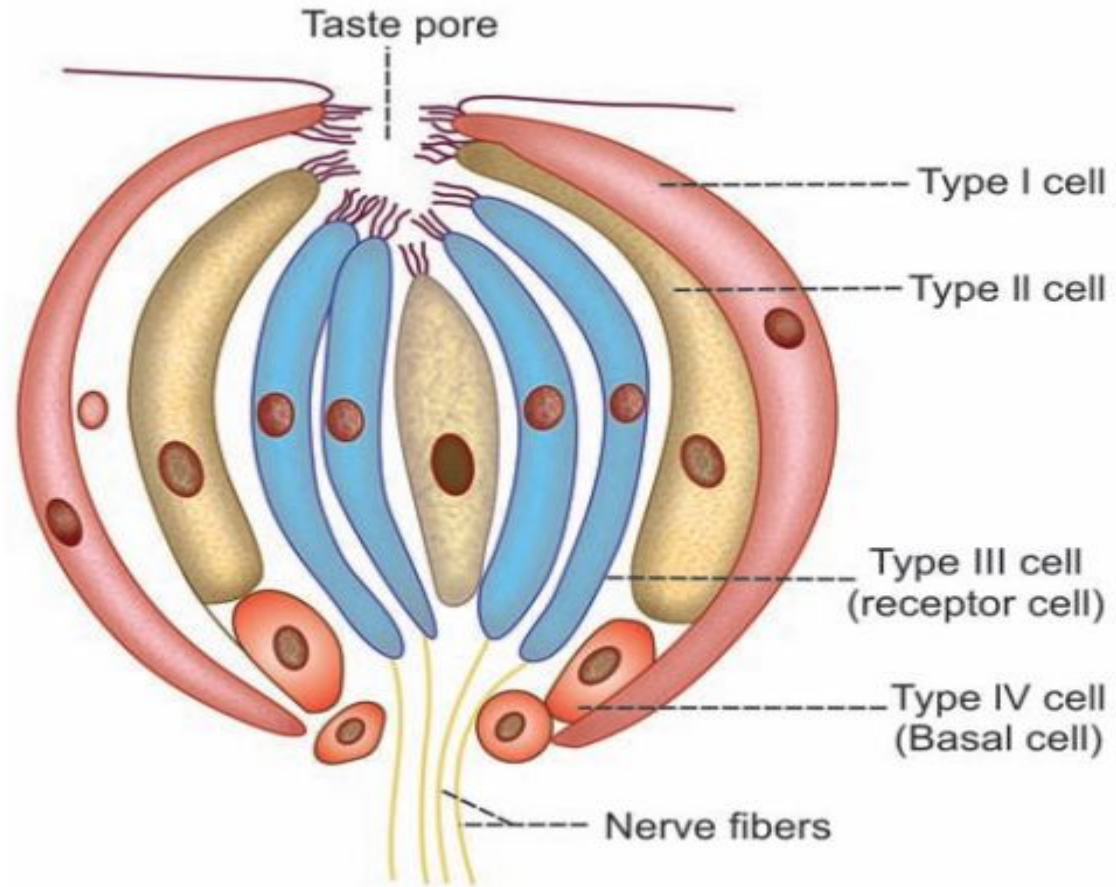
Taste buds contain the receptors for taste



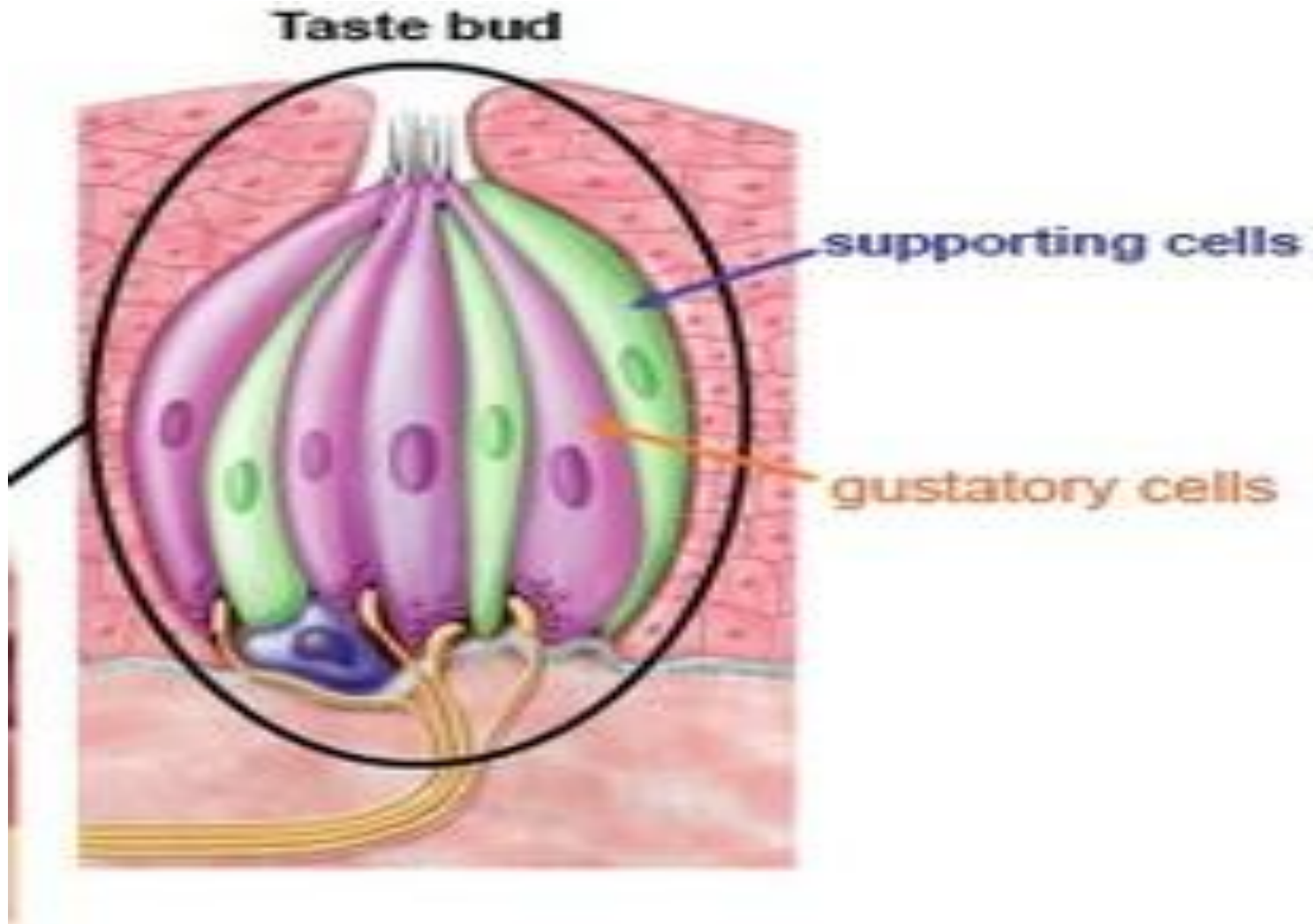
Structure of taste buds

- ***Each taste bud has 40 cells which are modified epithelial cells***
 - ***The cells are divided into four groups:***
 - Type 1 cells or sustentacular cells***
 - Type 2 cells***
 - Type 3 cells***
 - Type 4 cells or basal cells***
- Type 1 & 4 are supporting cells. Type 3 are taste cells. Function of type 2 are unknown.***

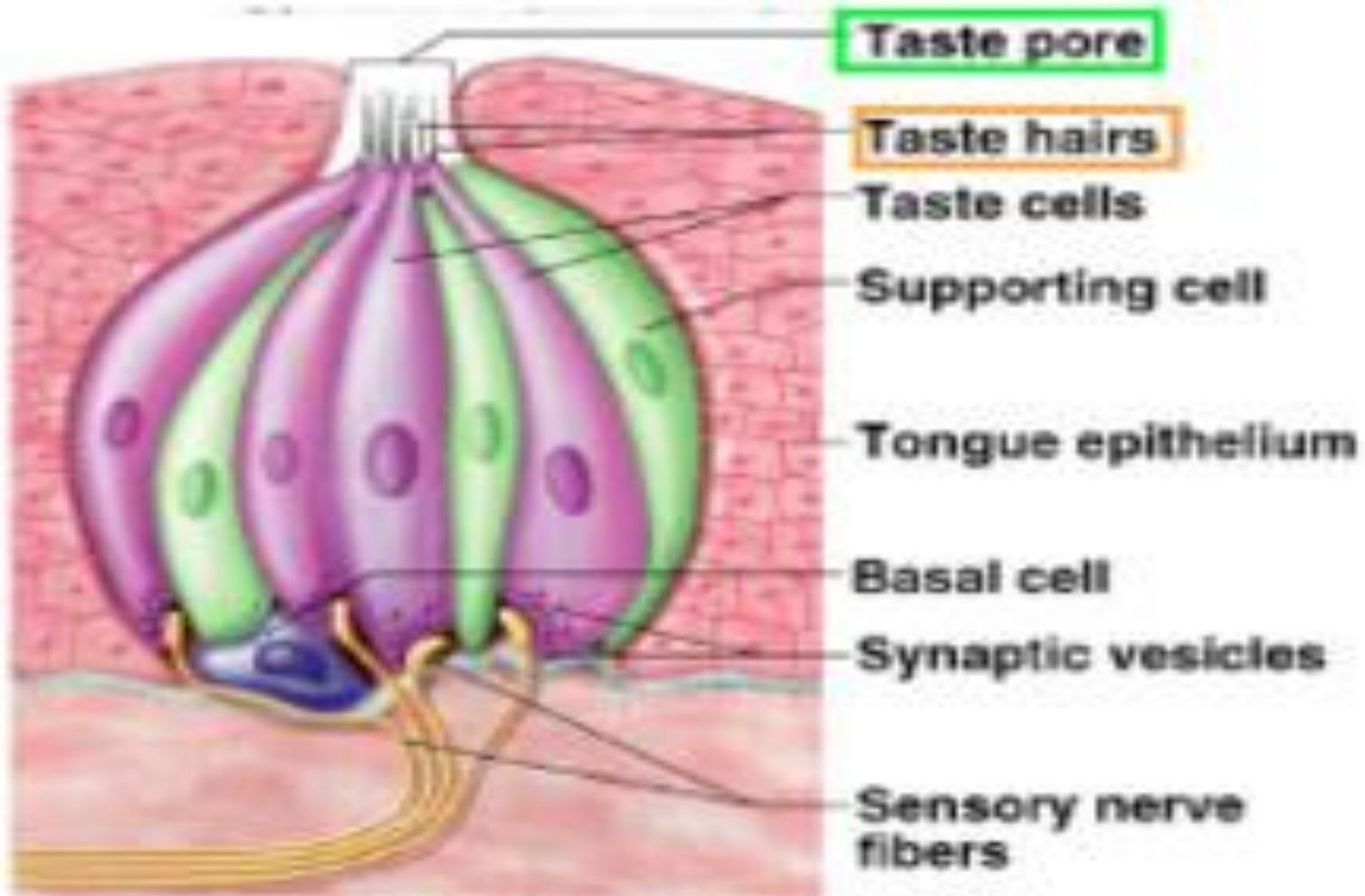
TASTEBUD



“The supporting cells & The gustatory (taste) cells”



Taste pore & taste hairs



Location of Taste buds

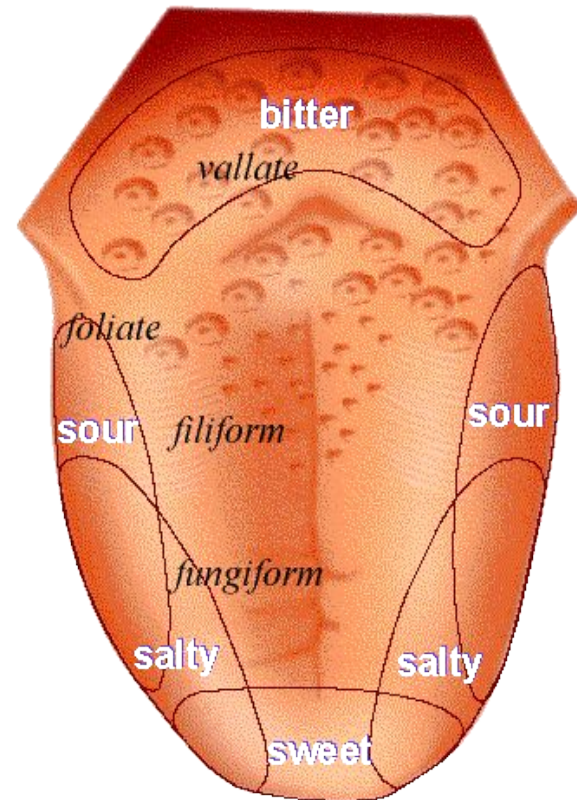
- ***They are located on the papillae ,which are present on***
- ***the upper surface of the tongue,***
- ***soft palate,***
- ***upper esophagus and***
- ***epiglottis***

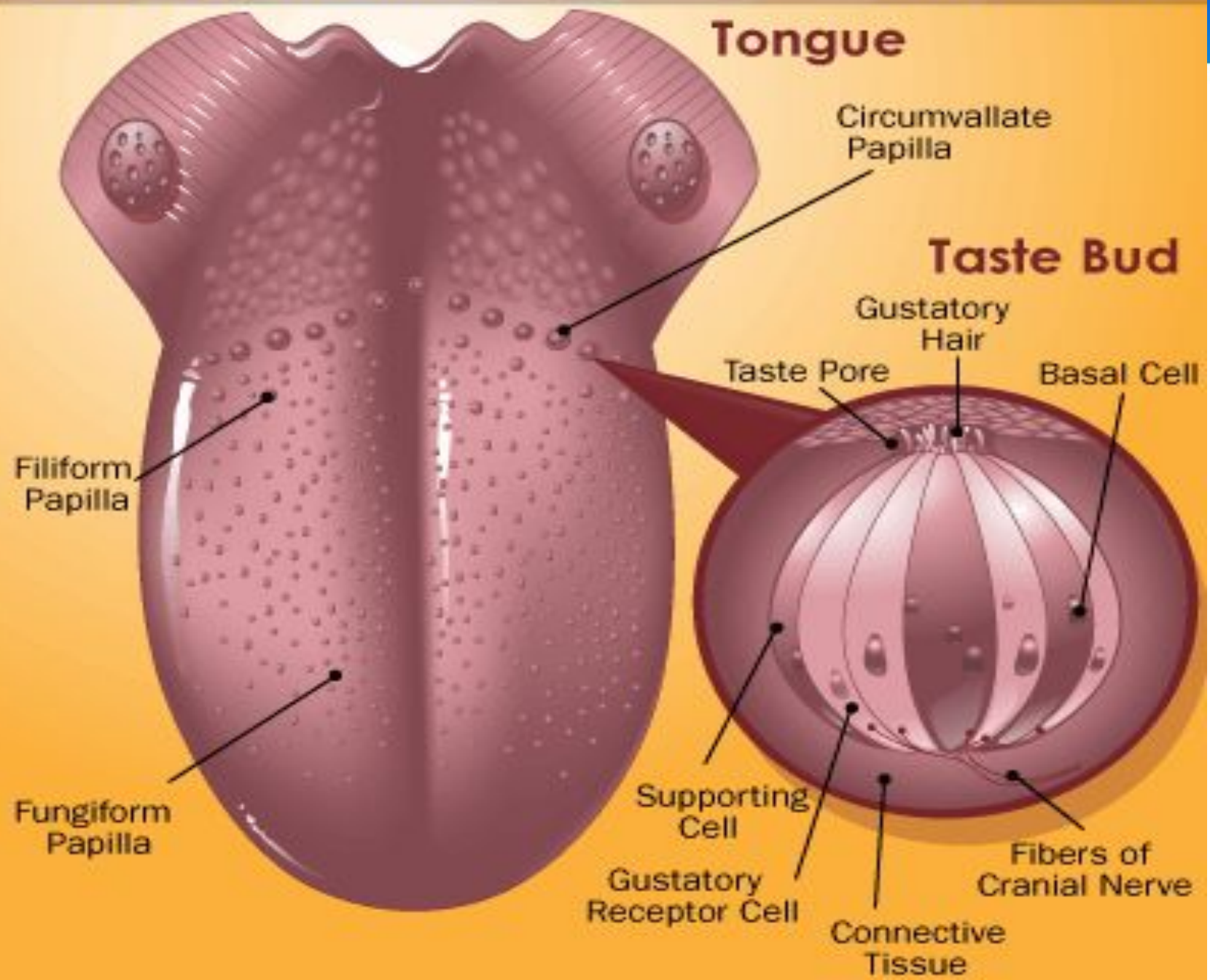
(On average, the human tongue has 2,000–8,000 taste buds.)

Types of papillae

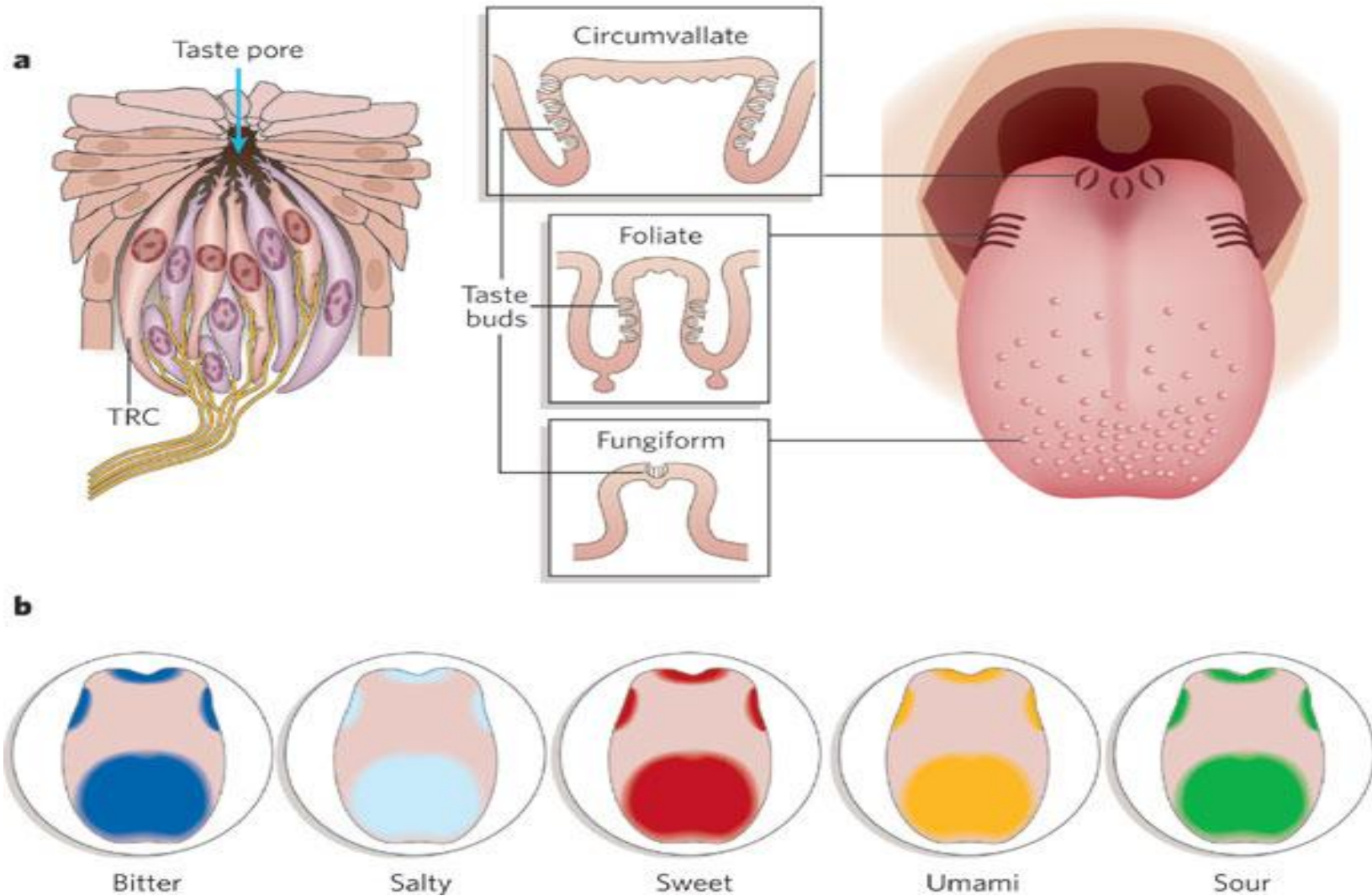
- ***The majority of taste buds on the tongue sit on raised protrusions of the tongue surface called papillae.***
- ***There are four types of papillae present in the human tongue.***

***# Fungiform papillae,
Filiform papillae,
Foliate papillae,
Circumvallate papillae.***





Types of papillae



Fungiform papillae

- ***These are slightly mushroom-shaped if looked at in longitudinal section.***
- ***These are present mostly at the apex (tip) of the tongue, as well as at the sides.***
- ***Innervated by facial nerve.***

Filiform papillae

- *These are thin.*
- *Long papillae .*
- *"V"-shaped cones*
- *That don't contain taste buds but are the most numerous.*
- *These papillae are mechanical and not involved in gustation.*
- *They are characterized by increased keratinization.*

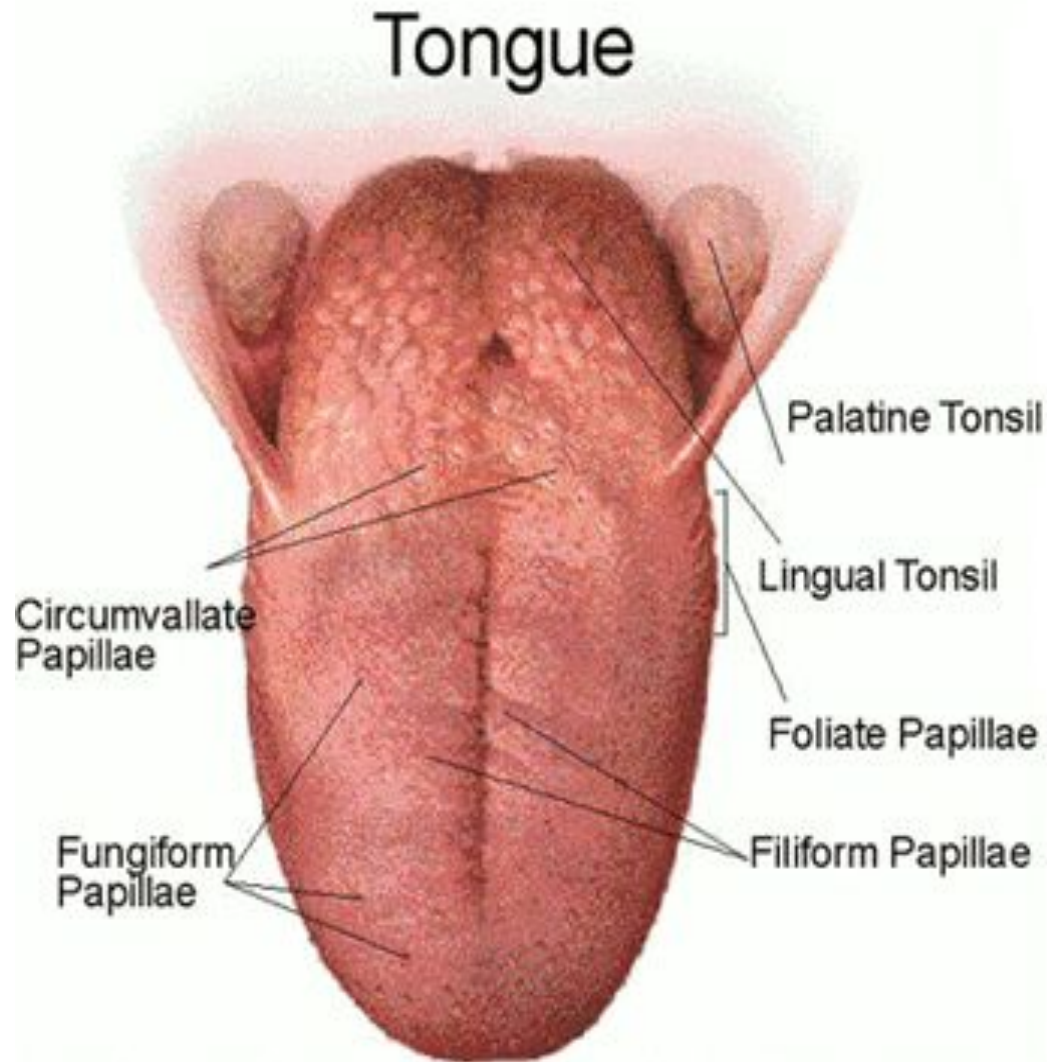
Foliate papillae

- *These are ridges and grooves towards the posterior part of the tongue found on lateral margins.*
- *Innervated by facial nerve (anterior papillae) and glossopharyngeal nerve (posterior papillae).*

Circumvallate papillae

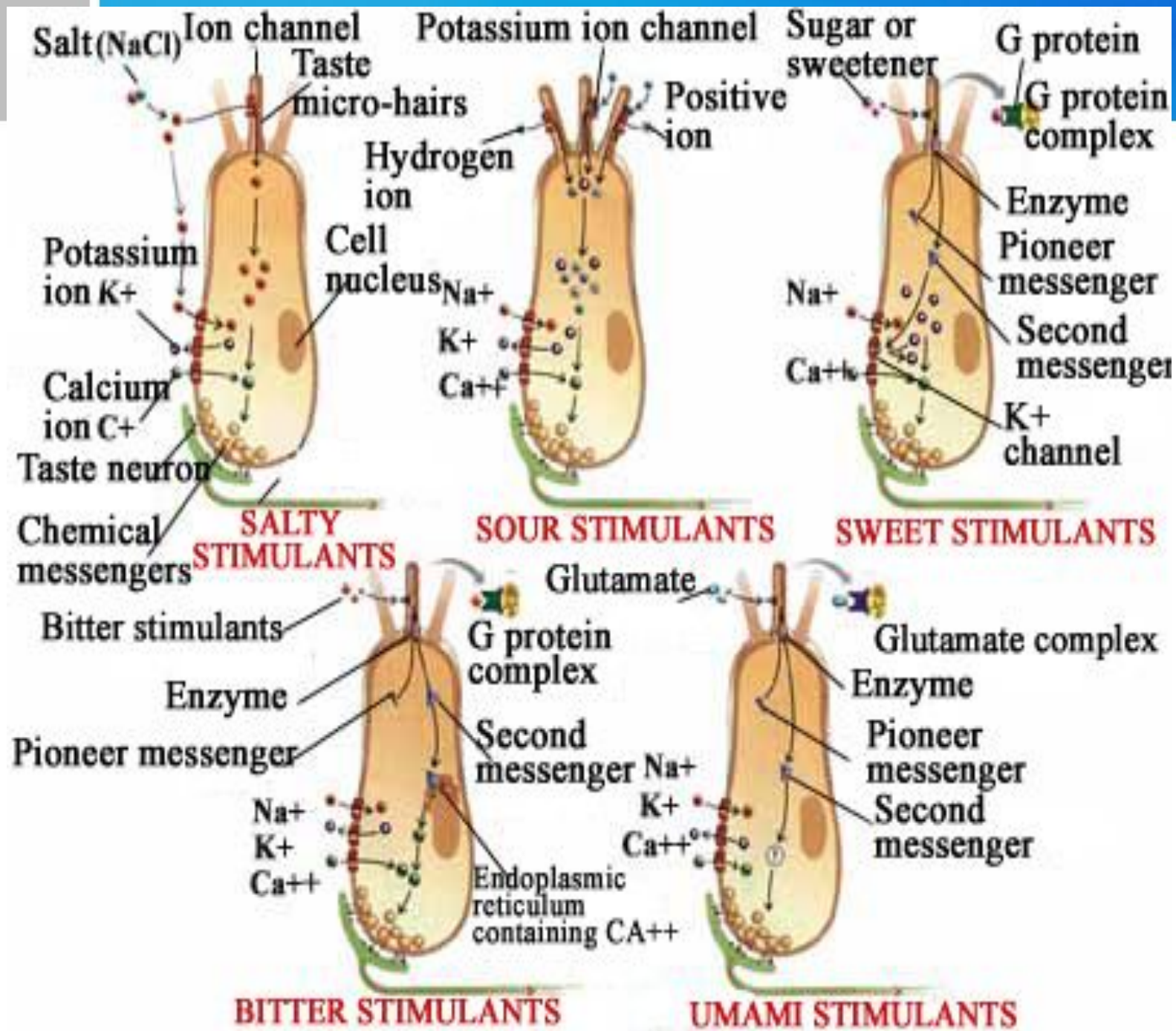
- *There are only about 10 to 14 of these papillae on most people.*
- *They are present at the back of the oral part of the tongue.*
- *They are arranged in a circular-shaped row just in front of the sulcus terminalis of the tongue.*
- *They are associated with ducts of Von Ebner's glands.*
- *Are innervated by the glossopharyngeal nerve.*

Types of papillae



Mechanism Of Stimulation Of Taste Receptors

- *Sweet, Bitter, and Savory (Umami), work with a signal through a **G protein-coupled receptor**.*
- *Salty and Sour, work with **ion channels**.*



Threshold for taste sensation

Sweet taste	sugar	1 in 200 dilution
Salt taste	Sodium chloride	1 in 400 dilution
Sour taste	Hcl	1 in 15,000 dilution
Bitter taste	quinine	1 in 2000,000 dilution

PATHWAY FOR TASTE

RECEPTORS

Receptors for taste sensation are the type III cells of taste buds. Each taste bud is innervated by about 50 sensory nerve fibers and each nerve fiber supplies at least five taste buds through its terminals.

FIRST ORDER NEURON

First order neurons of taste pathway are in the nuclei of three different cranial nerves, situated in medulla oblongata.

Dendrites of the neurons are distributed to the taste buds.

After arising from taste buds, the fibers reach the cranial nerve nuclei by running along the following nerves.

1. **Chorda tympani fibers of facial nerve**, which run from anterior two third of tongue
2. **Glossopharyngeal nerve fibers**, which run from posterior one third of the tongue
3. **Vagal fibers**, which run from taste buds in other regions.

Axons from first order neurons in the nuclei of these nerves run together in medulla oblongata and terminate in the nucleus of tractus solitarius.

□ **SECOND ORDER NEURON**

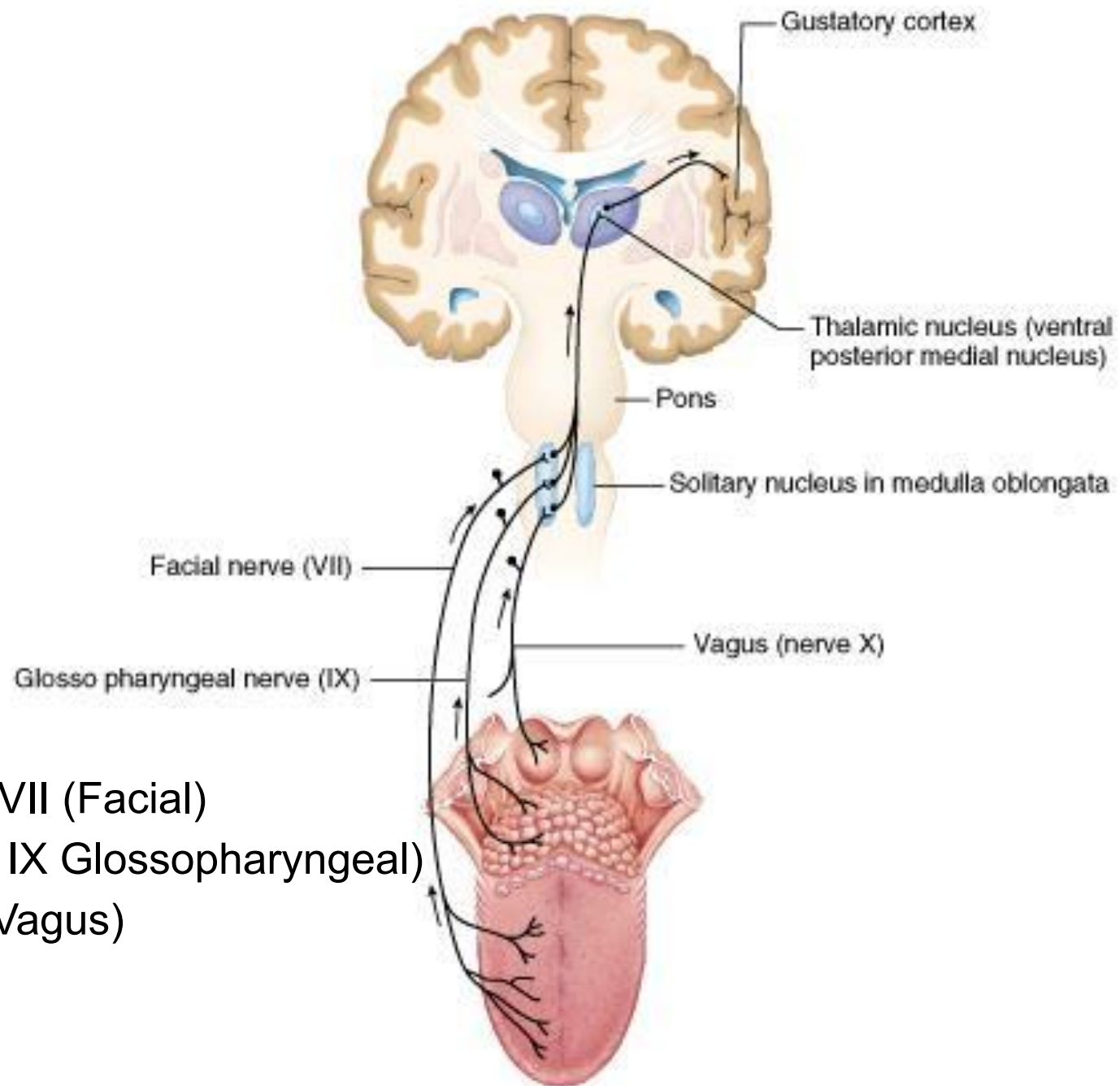
Second order neurons are in the nucleus of tractus solitarius. Axons of second order neurons run through medial lemniscus and terminate in posteroventral nucleus of thalamus.

□ **THIRD ORDER NEURON**

Third order neurons are in the posteroventral nucleus of thalamus. Axons from third order neurons project into parietal lobe of the cerebral cortex.

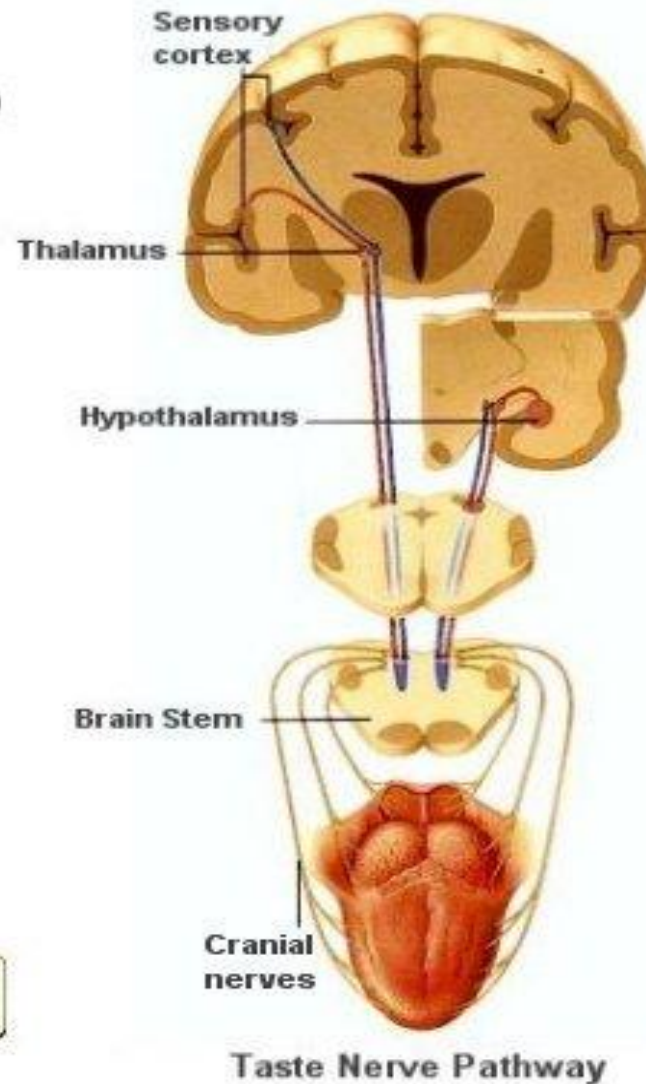
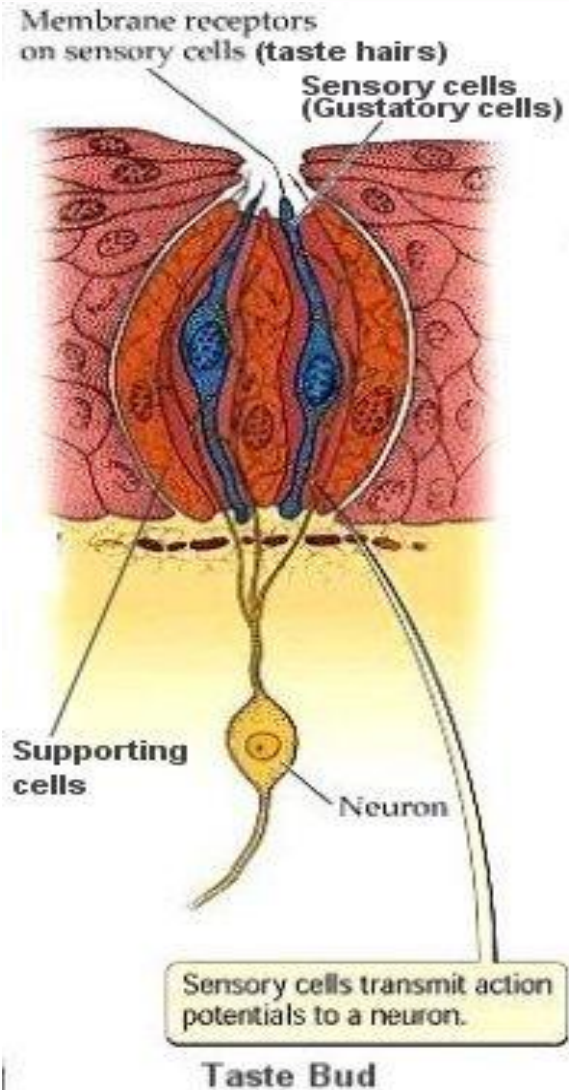
□ **TASTE CENTER**

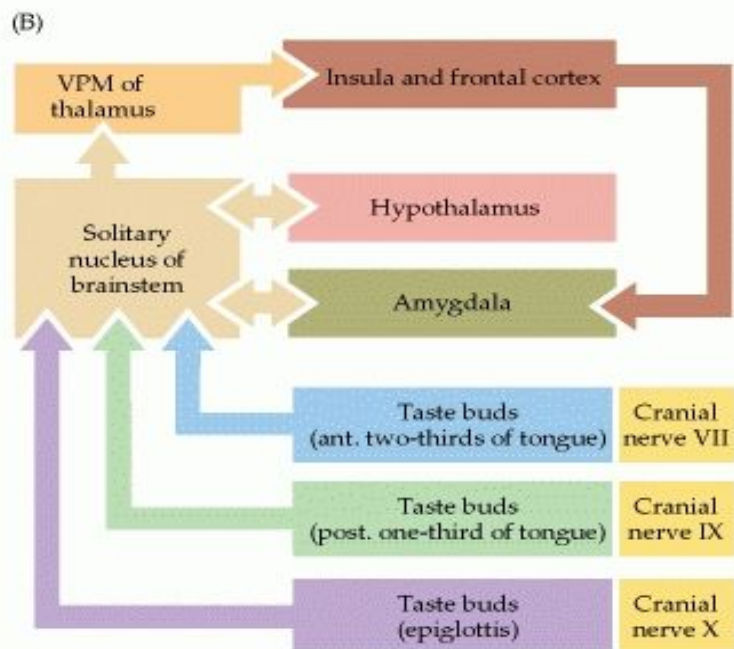
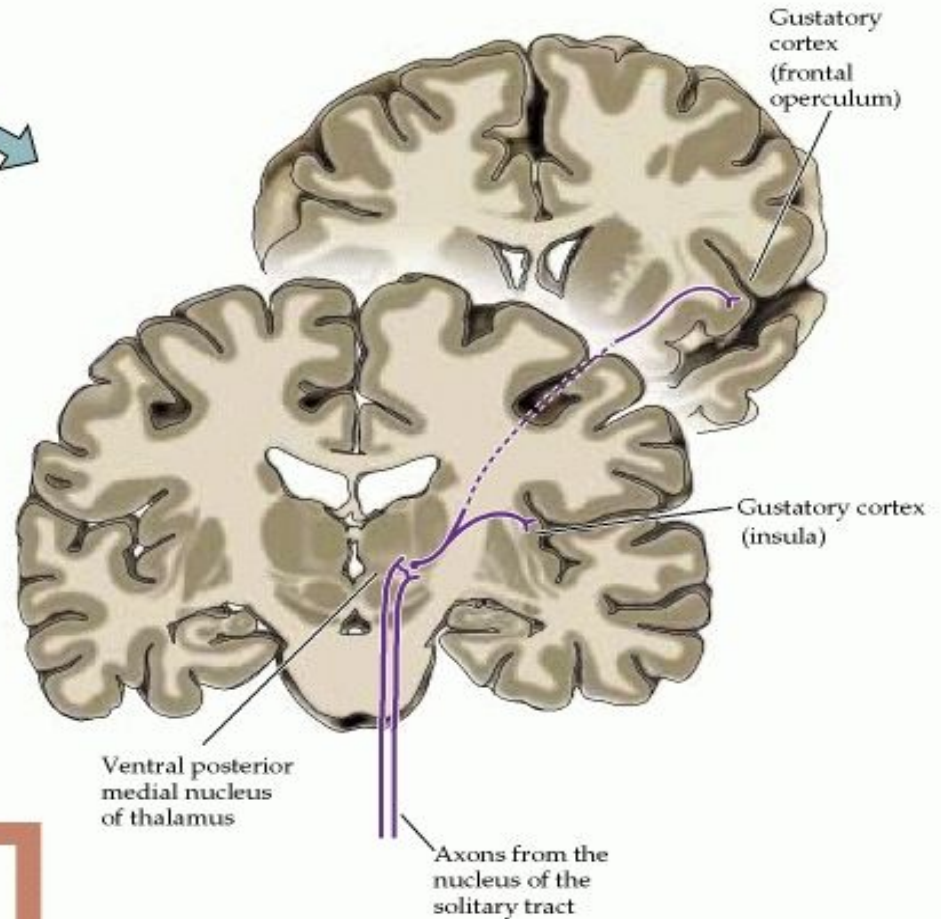
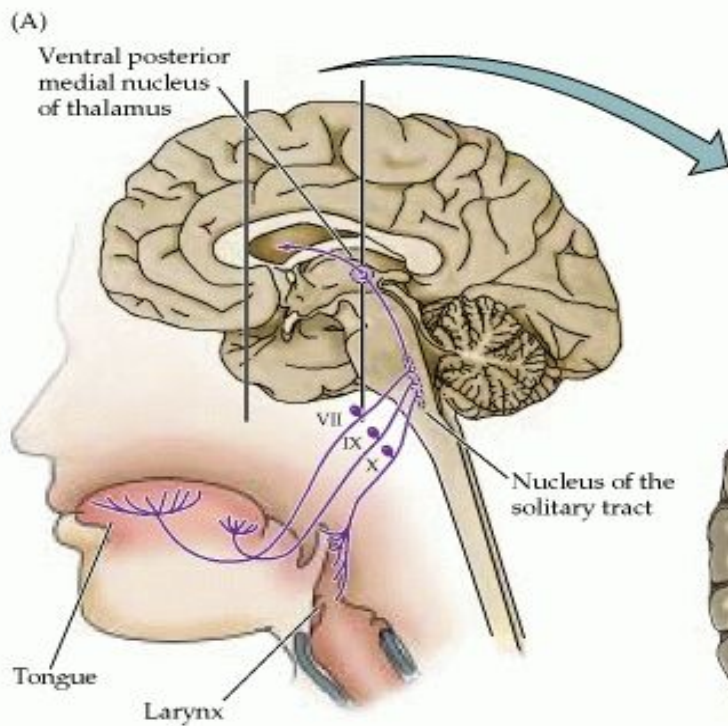
Center for taste sensation is in opercular insular cortex, i.e. in the lower part of postcentral gyrus, which receives cutaneous sensations from face. Thus, the taste fibers do not have an independent cortical projection.



Anterior 2/3 tongue: VII (Facial)
Posterior 1/3 tongue: IX Glossopharyngeal)
Pharynx: X (Vagus)

TASTE PATHWAY





TASTE TRANSDUCTION

- Taste transduction is the process by which taste receptor converts chemical energy into action potentials in the taste nerve fiber.
- Receptors of taste sensation are chemoreceptors, which are stimulated by substances dissolved in mouth by saliva.
- The dissolved substances act on microvilli of taste receptors exposed in the taste pore. It causes the development of receptor potential in the receptor cells. This in turn, is responsible for the generation of action potential in the sensory neurons

Taste Receptor

Generally, taste receptor is a G-protein coupled receptor (GPCR). It is also called G protein gustducin. However, several other receptors are also involved in taste sensation. Transduction mechanism is different in each taste receptor cells

Common Taste Disorders

- AGUSIA
- HYPOGEUSIA
- DYSGEUSIA

- **Ageusia** is the loss of taste functions of the tongue, particularly the inability to detect sweetness, sourness, bitterness, saltiness, and umami.

- **Causes**

- ✓ Neurological damage
- ✓ Deficiency of vitamin B3 (niacin) and zinc
- ✓ Cushing's syndrome, hypothyroidism and diabetes mellitus,
- ✓ penicillamine, cisplatin, ACE inhibitors, and other drugs
- ✓ glossitis
- ✓ tobacco
- ✓ aging
- ✓ anxiety disorder,
- ✓ cancer

Hypogeusia

- **Hypogeusia** is a reduced ability to taste things (to taste sweet, sour, bitter, or salty substances).
- Causes of hypogeusia include the chemotherapy, drug bleomycin, and antitumor antibiotic

Dysgeusia

- **Dysgeusia** is the distortion of the sense of taste
- The alterations in the sense of taste, usually a metallic taste, and sometimes smell are the only symptoms

Causes

- ❖ Chemotherapy
- ❖ Distortion of taste buds
- ❖ Zinc deficiency